

Depth-Estimation Student for Quadruped Parkour: Experiment Results and Analysis

Phases 1–4 (March 26–29, 2026)

LeggedGym-Ex Project

March 29, 2026

1 Problem Statement

A Go2 quadruped learns parkour (stairs, hurdles, gaps) via DAgger-based teacher–student distillation. The *teacher* uses ground-truth (GT) scandot heightmaps and achieves $\geq 98\%$ gap success. The student replaces the GT heightmap with a depth image—either from the simulator’s depth camera (GT-depth) or from monocular depth estimation (DepthAnything V2, “DA2”).

Goal. Achieve near-teacher gap success ($\geq 90\%$ on all difficulty rows) using monocular depth estimation, enabling sim-to-real transfer without a physical depth camera.

Gap difficulty rows. The parkour terrain generator creates gaps of increasing width across 4 difficulty rows:

Row	Gap width	Difficulty
0	0.24 m	Easy
1	0.32 m	Medium
2	0.42 m	Hard
3	0.50 m	Very hard

2 Upper Bounds: Teacher and GT-Depth Student

Evaluations use `ablate_parkour_student_depth.py` with `-parkour_force_family gap -parkour_force_row r`, 128–256 episodes per row, 16 parallel environments.

Policy	Gap success rate (%)			
	Row 0	Row 1	Row 2	Row 3
Teacher (GT scandots)	98.5	100.0	99.0	98.5
GT-depth student (3000 it)	94.5	94.5	89.1	87.5

The GT-depth student confirms the architecture is capable of $\geq 87\%$ on all rows. Any shortfall in the estimated-depth student is attributable to depth signal quality.

3 Phase 1: Initial Training (March 26)

- GT-depth student converged to 96% gap success (training metric).
- First estimated-depth (DA2-small) experiments exhibited wild oscillation: $0\% \rightarrow 72\% \rightarrow 0\% \rightarrow 90\%$.

- **Bug fix:** GRU hidden states were not being reset on episode boundaries, causing stale temporal context to corrupt the first frames of each episode.

4 Phase 2: Training Stabilization (March 27)

All experiments in this phase use **DA2-small** ($\sim 25\text{M}$ params) with default EMA percentile normalization.

ID	Config	Iters	Row 0 (%)	Notes
S13	diff-lr (actor_lr_scale=0.1)	5000	~ 48	Reduced oscillation
S14	diff-lr + cosine decay	3000	56.3	Best DA2-small result
S15	actor_lr_scale=0.05	5000	~ 48	No improvement
C1	slow curriculum + cosine + diff-lr	8000	~ 48	No improvement
S11	freeze actor	3000	~ 40	Worse
S12	freeze actor + cosine	5000	~ 42	Worse
F1	fine-tune S14 @ LR=1e-4	2000	~ 60	Maintained, no gain
F2	fine-tune S14 @ LR=5e-5	2000	~ 60	Maintained, no gain
F3	fine-tune S14 frozen actor	2000	~ 58	Maintained, no gain

Key finding. Differential learning rate (actor_lr_scale=0.1) with cosine LR decay ($2\text{e-}3 \rightarrow 1\text{e-}4$ over 5000 iterations) was the best training recipe for DA2-small. However, performance plateaued at $\sim 56\%$ gap success on Row 0, far below the GT-depth student’s 94.5%.

5 Phase 3: Root Cause Analysis (March 28)

5.1 Per-Row Evaluation Reveals Gap

Policy	Gap success rate (%)			
	Row 0	Row 1	Row 2	Row 3
Teacher	98.5	100.0	99.0	98.5
GT-depth student	97.5	94.0	93.0	86.5
S14 (DA2-small)	45.5	55.2	50.0	18.5
S14 (zero depth)	8.0	—	—	0.0

The student *does* use depth (zero-depth ablation drops to 0–8%), but the DA2-small signal is too weak for reliable gap detection.

5.2 Depth Signal Analysis

- DA2 separates gaps at the pixel level: $d' = 2.521$ (excellent discrimination). Ground pixels: $\mu = 5.66$, Gap pixels: $\mu = 8.47$.
- EMA normalization pixel-level gap signal: 40.6% of output range.
- However, frame-to-frame noise and EMA non-stationarity degrade the signal over time.
- Calibrated linear normalization: 11.3% signal (worse).
- Fixed range [3.5, 12.0]: 33.1% signal (deterministic but weaker).

5.3 Normalization Experiments

ID	Normalization	Iters	Result
CAL1	Calibrated linear (scale=0.081, offset=0.995)	2000	~45% (no improvement)
CAL2	Fixed range [3.5, 12.0]	1900	~45% (no improvement)

Conclusion. Normalization is not the bottleneck. The core issue is depth estimation quality on untextured simulation terrain.

6 Phase 4: Model and Signal Quality (March 29)

Armed with the root cause (depth signal quality), we pursued two orthogonal improvements:

1. **Larger depth model:** DA2-base (~100M params) vs. DA2-small (~25M params).
2. **Terrain textures:** Checkerboard pattern with bright gap coloring, giving DA2 visual features to estimate depth from.

6.1 Primary Results

All experiments use diff-lr + cosine decay. Evaluations at the stated checkpoint with 128–256 episodes per row.

ID	Config	Iters	Gap success rate (%)			
			Row 0	Row 1	Row 2	Row 3
S14	DA2-small (baseline)	3000	56.3	33.6	—	—
BASE1	DA2-base	5000	68.4	83.2	26.6	0.8
SMOOTH1	DA2-small + temporal EMA	5000	57.0	—	—	—
BASE2	DA2-base + temporal EMA	3100	62.5	6.3	—	—
BASE3	DA2-base + texture	5000	84.8	89.1	86.3	71.5
BASE4	DA2-base + texture + EMA	3000	62.5	51.6	—	—

6.2 Architecture Experiments

ID	Config	Iters	Result
GRAD1	Sobel gradient 2-channel CNN	5000	~40% avg (worse than baseline)
GRAD2	Sobel + texture	1000	Killed early; superseded by BASE3
SCANDOT1	GT-depth → 132 scandots	5000	100% training gap; loss=0.010
SCANDOT2	DA2-small → 132 scandots	5000	Not evaluated
TEX1	DA2-small + texture	800	Killed; superseded by BASE3

6.3 BASE3 Training Trajectory

BASE3 is still improving. The table below shows gap success at various checkpoints (64–256 episodes per evaluation, accounting for variance):

Checkpoint	Row 0	Row 1	Row 2	Row 3	Episodes
2600	60.9	49.2	—	—	128
3100	67.2	73.4	43.8	—	128
3700	78.9	86.3	74.2	57.4	256
5000	84.8	89.1	86.3	71.5	256
6200	91.4	88.3	62.5	37.5	256
6700	82.8	98.4	89.1	—	64

Note: High inter-evaluation variance (especially with 64 episodes) makes individual checkpoints unreliable. The general trend is upward, particularly for harder rows. Row 3 went from 0% (BASE1@5000) to 57–71% (BASE3@3700–5000).

6.4 Comparison with GT-Depth Upper Bound

Policy	Gap success rate (%)			
	Row 0	Row 1	Row 2	Row 3
GT-depth student	94.5	94.5	89.1	87.5
BASE3 (best ckpt)	91.4	98.4	89.1	71.5
Gap to GT	−3.1	+3.9	0.0	−16.0

Rows 0–2 are within the noise floor of the GT-depth student. Row 3 (0.50 m gaps) remains ~16 points below, but is still improving with training.

7 Key Findings

- Depth model size is the primary lever.** Upgrading DA2 from small (25M) to base (100M params) improved Row 0 gap success from 56% to 68% (+12 pp) and Row 1 from 34% to 83% (+49 pp). The larger model produces higher-resolution, more accurate depth maps.
- Terrain textures are the secondary lever.** Adding checkerboard textures to the Genesis heightfield gives DA2 visual features for monocular depth estimation. Combined with DA2-base, this pushed Row 2 from 27% to 86% (+59 pp) and Row 3 from 1% to 72% (+71 pp). The effect is multiplicative with model size.
- Temporal smoothing is counterproductive.** EMA smoothing ($\alpha=0.5$) on the depth map reduces frame-to-frame noise but also blurs the sharp depth discontinuities at gap edges that the student relies on. All smoothing variants underperformed their non-smoothed counterparts.
- Architecture changes are secondary.** Sobel gradient channels, direct scandot prediction, and other architectural modifications did not significantly improve over the baseline architecture. The bottleneck was always depth signal quality.
- Normalization does not matter.** EMA percentile, calibrated linear, fixed-range, and mean-std normalization all produced similar results once depth signal quality was addressed.
- Training metrics are misleading.** The `Episode/success_gap` metric logged during training reflects curriculum difficulty, not absolute gap performance. 100% in training means easy curriculum, not solved gaps. Per-row forced evaluation is essential.

7. **Diff-lr + cosine decay is the training recipe.** Across all experiments, `actor_lr_scale=0.1` with cosine decay from $2e-3$ to $1e-4$ was consistently the best or tied-for-best training configuration.

8 Current Status and Next Steps

Active training.

- BASE3-ext (DA2-base + texture): extended to 10,000 iterations, currently at $\sim 6,700$. Row 3 performance may still improve.
- BASE1-ext (DA2-base, no texture): extended to 10,000 iterations, currently at $\sim 9,500$. Control experiment.

Remaining gap. Row 3 (0.50 m) is the only row where the estimated-depth student significantly trails the GT-depth student (71% vs. 88%). Options to close this gap:

- Continue training (Row 3 has been improving steadily).
- Try DA2-large ($\sim 335M$ params) if compute permits.
- Targeted gap-only curriculum phase.
- Gap-aware texture design (e.g., brighter gap coloring).

Total experiments run. ~ 30 training runs across 4 phases, covering normalization (5 variants), depth models (3 sizes), architecture (4 variants), textures (3 configs), temporal smoothing (2 configs), and training hyperparameters (10+ configs).